Generative modelling of the bacterial J-domain family with an attention-parametrised Potts model

TP4 — AttentionDCA

September 30, 2026

Abstract

We fit a Potts model of the bacterial J-domain family ($L = 63$, $q = 21$) whose coupling tensor is parametrised by a factored self-attention layer, train it by maximising the reweighted pseudo-likelihood, and sample new sequences from the learned conditionals. Since the supplied train/test split repeats 79% of its sequences verbatim across the two sides, we work throughout on a sequence-disjoint split rebuilt from the full 36 713-sequence pool. On it the model reaches a held-out pseudo-likelihood 11% above its training value and recovers 1.00 precision on the top $L/2$ structural contacts. Sequences generated by the pseudo-likelihood sampler reproduce the natural one-site frequencies ($r = 0.990$) and connected two-site correlations ($r = 0.871$, against a train-vs-test ceiling of 0.944), are new (mean distance 0.42 to the closest training sequence, against 0.18 for natural held-out sequences) and as diverse as natural sequences (0.68 vs. 0.60 mean pairwise distance). Chains decorrelate from their seed in 11 sweeps ($\tau_{\mathrm{int}} = 6.5$). The one systematic failure is a minor HscB-like sub-family carrying 14.1% of the natural weight but only 5.1% of the samples; restricting the comparison to the modes the sampler does populate raises the correlation agreement to 0.918 against a 0.941 ceiling. That sub-family is not mis-modelled but mis-weighted: its conditionals are fitted as well as any other cluster's, while its energy sits 14 units too high — pseudo-likelihood constrains conditionals, never the relative masses of the basins, so the model learns each sub-family but not the prior over them.

1 Introduction

Direct Coupling Analysis models a protein family as a Potts distribution over aligned sequences,

$$P(\boldsymbol{x}) = \tfrac{1}{Z} e^{-E(\boldsymbol{x})}, \qquad E(\boldsymbol{x}) = -\sum_{i<j} J_{ij}(x_i, x_j), \quad (1)$$

whose couplings $J_{ij}(a, b)$ encode the co-evolution between positions. Fitting a full coupling tensor means $\binom{L}{2}q^2$ parameters, which for $L = 63$, $q = 21$ is 861 273 numbers estimated from a few thousand effectively independent sequences.

AttentionDCA replaces that tensor by a factored attention layer. With H heads carrying query/key matrices $Q^h, K^h \in \mathbb{R}^{d\times L}$ and a value matrix $V^h \in \mathbb{R}^{q\times q}$,

$$A^h_{ij} = \mathrm{softmax}_j\left(\textstyle\sum_k Q^h_{ki} K^h_{kj}\right), \quad A^h_{ii} = 0, \quad (2)$$

$$J_{ij}(a, b) = \sum_{h=1}^{H} A^h_{ij}\, V^h(a, b), \quad (3)$$

which separates *which* positions interact (A^h) from the amino-acid chemistry of the interaction (V^h) and costs only $H(2dL + q^2)$ parameters. Following the original formulation no explicit fields are used: conservation is absorbed into the couplings.

This work has three aims: to train that model on the J-domain family by pseudo-likelihood, to sample new sequences from it and to ask, quantitatively, in what respects those sequences do and do not look like natural ones.

2 Methods

Pseudo-likelihood training. Z in Eq. (1) is intractable, so we maximise the pseudo-likelihood built from the exact single-site conditionals

$$p(x_i = a \mid \boldsymbol{x}_{\setminus i}) = \frac{\exp \sum_{j\neq i} J_{ij}(a, x_j)}{\sum_c \exp \sum_{j\neq i} J_{ij}(c, x_j)}, \quad (4)$$

minimising $\mathcal{L} = -\sum_m w_m \sum_i \log p(x^m_i \mid \boldsymbol{x}^m_{\setminus i}) + \lambda \sum_{i\neq j,a,b} J_{ij}(a, b)^2$ with Adam (lr $5 \cdot 10^{-3}$, batches of 1000), a 10% validation split and early stopping on the validation pseudo-likelihood.

Reweighting. Natural alignments over-represent well-sampled clades, so each sequence carries $w_m = 1/|\{n : d_H(\boldsymbol{x}^m, \boldsymbol{x}^n) < \theta L\}|$ with $\theta = \min(0.5, 0.38 \cdot 0.32/\langle\text{fraction identity}\rangle)$, giving $M_{\mathrm{eff}} = \sum_m w_m$. The same weights are used for every natural statistic reported below; generated sets are weighted uniformly.

Sampling. New sequences are drawn with the learned conditionals: one *sweep* visits the L sites in random order and resamples each from Eq. (4) raised to an inverse temperature β. Since A^h is row-normalised, $J_{ij} \neq J^\top_{ji}$ and the chain is only approximately Boltzmann; we therefore monitor both the symmetrised Hamiltonian $E(\boldsymbol{x}) = -\frac{1}{2}\sum_{i\neq j} J_{ij}(x_i, x_j)$ and the pseudo-log-likelihood $\log \mathrm{PL}(\boldsymbol{x}) = \sum_i \log p(x_i \mid \boldsymbol{x}_{\setminus i})$. All chains are advanced simultaneously on the GPU, which costs $\sim$0.3 μs per chain-update against 0.6–28 ms for the original per-site Python loop.

Metrics. We compare weighted one- and two-site statistics $f_i(a)$, $f_{ij}(a, b)$ and $C_{ij}(a, b) = f_{ij}(a, b) - f_i(a)f_j(b)$ entry-wise, using the Pearson r (tightness of the relation) and the least-squares slope of generated on natural (amplitude; the two are linked by slope $= r\,\sigma_{\mathrm{gen}}/\sigma_{\mathrm{nat}}$). Novelty and diversity use the normalised Hamming distance $d(\boldsymbol{x}, \boldsymbol{y}) = \frac{1}{L}\sum_i (1 - \delta_{x_i, y_i})$.

Two references bracket every number: *train vs. test*, the finite-sample ceiling, and an *independent-site* model sampled from the training f_i, which matches f_i by construction and has $C_{ij} \equiv 0$.

Data and split. An audit of the alignments shipped with the project showed that its train/test pair is species-disjoint but not sequence-disjoint: 78.8% of the test sequences are byte-identical to a training sequence, because the same J-domain recurs across species. We therefore build the split from the full 36 713-sequence pool by clustering greedily (CD-HIT style, 80% identity, 10 942 clusters) and assigning whole clusters to one side, removing afterwards only sequences identical to a training one. This gives 29 312 training and 7 401 test sequences with no exact copies (median distance to the nearest training sequence 0.175). Test sequences that merely *resemble* training ones are kept — discarding them would leave only the isolated corners of the family and misrepresent the density the generative model must reproduce. The companion report documents the audit in full.

Units. Log-likelihoods use the natural logarithm and are quoted in *nats* (1 nat $= 1/\ln 2 \approx 1.44$ bits): a difference of Δ nats means one model assigns the data e^{Δ} times more probability, so 0.7 nats is a factor of two.

3 Results

3.1 Training and structural validation

A grid over H, d and λ scored by validation pseudo-likelihood (Table 1) selects $H = 128$, $d = 10$, $\lambda = 10^{-3}$ (217 728 parameters). Performance improves with head count up to ∼128 heads and then saturates: doubling to 256 buys 0.3 nats per sequence for twice the parameters.

The fitted model attains 71.10 nats per sequence on training data and 78.72 on the held-out set (Fig. 1). Because that test set shares no sequence with the training set, the 11% difference is a genuine generalisation gap. On the supplied split the same quantity is almost flat (75.70 vs. 85.61), which is exactly what one expects when most "held-out" sequences were in fact seen.

That the couplings are physically meaningful, and not merely predictive, is confirmed independently of any generation: after symmetrisation, zero-sum gauge, Frobenius norm and average product correction, the top-ranked long-range pairs ($|i - j| > 4$) are true contacts (< 8 Å) with precision 1.00 at $L/2$ pairs and 0.83 at L pairs, against a 0.21 base rate.

3.2 The sampler mixes, and needs a temperature

Chains started from natural training sequences and from uniform random sequences converge to the same stationary energy (-126.6 vs. -126.6 at $\beta = 1$), so the stationary state does not depend on the initialisation (Fig. 2). The distance to the seed reaches 90% of its plateau after 11 sweeps, and the plateau (0.66) matches the mean test–test distance: the chain forgets its starting sequence entirely. The integrated autocorrelation time of the energy is $\tau_{\mathrm{int}} = 6.5$ sweeps. We generate 10 000 sequences per temperature, each the final state of an independent chain run for 500 sweeps ($\approx 40\,\tau_{\mathrm{int}}$). No chain ends in a degenerate gap-only state.

H	d	λ	params	best epoch	val. $-\log$ PL
256	10	10^{-3}	435 456	187	85.33
128	10	10^{-3}	217 728	387	85.61
64	10	10^{-3}	108 864	709	86.73
64	10	10^{-4}	108 864	168	87.34
64	23	10^{-4}	213 696	293	87.37
64	23	10^{-3}	213 696	654	87.63
32	10	10^{-4}	54 432	817	87.78
32	23	10^{-4}	106 848	697	88.17
32	23	10^{-3}	106 848	2061	88.22
32	10	10^{-3}	54 432	817	89.10
16	10	10^{-4}	27 216	973	89.74
16	23	10^{-4}	53 424	820	90.61
16	23	10^{-3}	53 424	2024	90.64
16	10	10^{-3}	27 216	1692	91.34

Table 1: Hyper-parameter grid sorted by validation $-\log$ PL per sequence. The chosen model is the second row.

Pseudo-likelihood constrains the conditionals but not the global normalisation, so $\beta = 1$ need not reproduce the energy scale of natural sequences. Figure 3 scans β: raising it deepens the energy and sharpens the correlations at the cost of diversity. We report both $\beta = 1$, the learned model as it stands, and $\beta^* = 1.05$, defined by matching the mean sample energy to the reweighted mean energy of the *training* set — a criterion that never touches the test set.

3.3 One- and two-site statistics

Site frequencies are reproduced almost exactly ($r = 0.990$ at $\beta = 1$, 0.992 at β^*), but so are they by the independent-site baseline, so the informative quantity is the connected correlation. There the model reaches $r = 0.871$ ($\beta = 1$) and 0.859 (β^*) against a finite-sample ceiling of 0.944 and a baseline of -0.005 (Fig. 4, Table 2).

The regression slope separates two different errors. Since slope $= r\,\sigma_{\mathrm{gen}}/\sigma_{\mathrm{nat}}$, it reports amplitude where r reports tightness. At $\beta = 1$ the slope is 0.78: the model identifies the right correlations but systematically *under-states* them, its samples being under-dispersed. At β^* it rises to 0.85, against a train-vs-test ceiling of 0.88.

That the ceiling is itself below 1 is regression dilution: the natural C_{ij} are estimated from only $M_{\mathrm{eff}} = 1280$ effective sequences, and noise in the predictor biases any least-squares slope towards zero. Correlating two independent halves of the test set and applying the Spearman–Brown formula puts the reliability of the natural C_{ij} at 0.94; dividing through by it gives disattenuated slopes of 0.83 ($\beta = 1$), 0.91 (β^*) and 0.94 (ceiling). Read against that ceiling rather than against 1, β^* recovers the correlation amplitude to within a few

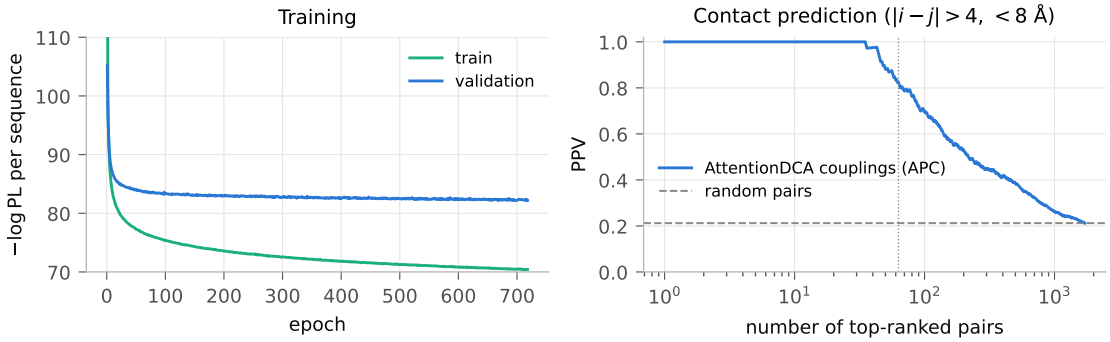


Figure 1: Left: weighted $-\log \mathrm{PL}$ per sequence. Right: contact-prediction precision of the ranked APC-corrected coupling norms; vertical line at L pairs.

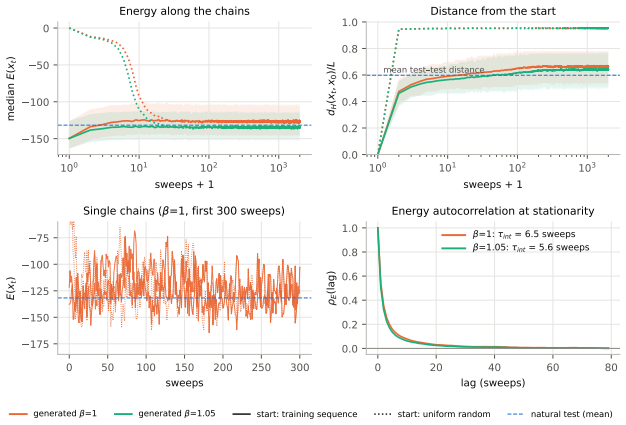


Figure 2: Pseudo-likelihood sampling. (a) median energy (band: 10–90% of chains from training starts); (b) Hamming distance to the starting sequence; (c) individual chains at $\beta = 1$; (d) chain-averaged energy autocorrelation at stationarity.

percent, while $\beta = 1$ falls short by about 10%. The ceiling does not reach 1 because clustering sends whole groups to one side, so train and test sample somewhat different regions of the family.

3.4 A missing sub-family, and what it costs

Projected on the first two principal components of the test alignment (Fig. 5), the family splits into two large clusters and one minor one. k-means with $k = 3$ (the elbow of the within-cluster inertia) makes this quantitative: writing s_k for a cluster's share of a set's weight, 2 clusters satisfy $s_k^{\mathrm{gen}}/s_k^{\mathrm{nat}} \geq 0.6$ at both temperatures and hold 85.9% of the natural test weight. The remaining cluster, an HscB-like group of J-domains, carries 14.1% of the natural weight but receives only 5.1% ($\beta = 1$) and 3.2% (β^*) of the samples. Roughly one chain in ten *starts* inside that cluster and leaves, so the sampler can reach it; the model simply assigns it too little stationary weight.

Table 3 says why, and the answer is that the model never learns the *priors* — the relative masses of the sub-families. On training data the pseudo-likelihood of the missed cluster is essentially that of the others (-71.8 vs. -71.1/-71.0 nats), so its local statistics are fitted just as well. Its energy is not: natural sequences of C0 score -119.5 (test) and -122.6 (train), about 14 units above the covered clusters (-134.0/-133.5, -134.5/-137.2). A basin's stationary weight follows its free energy, so that offset is what starves the cluster — a measured free-energy error of $\Delta F = +1.01$ nats, i.e. 0.36 of the natural weight, against $|\Delta F| \leq 0.23$ elsewhere. This is intrinsic to the objective: maximising Eq. (4) constrains local conditionals and never the global normalisation, so nothing in the loss asks the model how much total mass each sub-family should carry, and with no fields and no mixture component there is nowhere for such a prior to live. Reweighting compounds it by design, since it flattens exactly the clade-frequency signal that would carry that information.

This single defect accounts for most of the residual disagreement in the correlations. Repeating the comparison with *both* sets restricted to the covered clusters (Fig. 6, Table 4) raises the agreement from 0.871 to 0.918 ($\beta = 1$) and from 0.859 to 0.918 (β^*), while the train-vs-test ceiling barely moves (0.944 → 0.941). Inside the modes it populates, the model sits close to the finite-sampling limit; what it gets wrong is the relative weight of the modes themselves.

3.5 The sequences are new, and as diverse as natural ones

Novelty and diversity are read from three different nearest-neighbour distributions (Fig. 7, Table 5). Generated sequences sit at 0.42 ($\beta = 1$) and 0.39 (β^*) from the closest *training* sequence, whereas a genuine held-out natural sequence sits at only 0.18: the model is not reproducing its training set. All 10 000 samples are distinct (100.0% unique). Their internal spread, 0.68 at $\beta = 1$, matches the natural test–test spread of 0.60, and the generated-vs-test distribution lies on top of it, so the samples neither collapse onto a few modes nor scatter into regions the family does not occupy. Raising the temperature to β^* costs a little of this diversity (0.65), which is the price of the corrected correlation amplitude.

Consistently, the energy distribution of the samples at β^* overlays the natural one (Wasserstein distance 2.1, against 5.9 at $\beta = 1$ and 39.5 for the independent-site model).

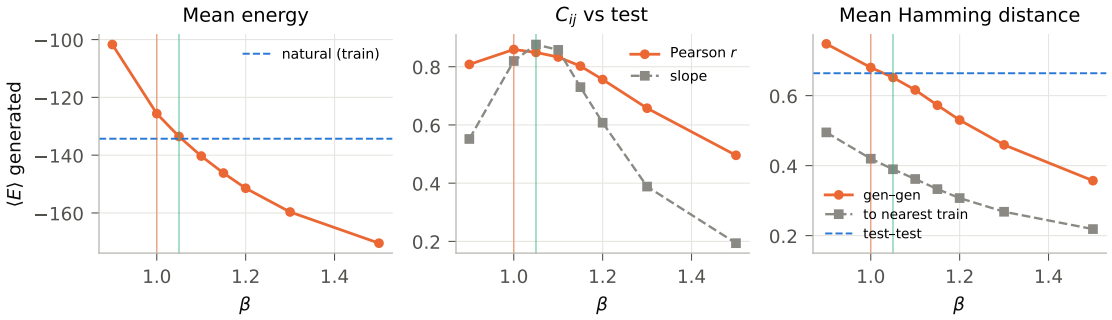


Figure 3: Inverse-temperature scan (chains started from training sequences). Vertical lines mark $\beta = 1$ and $\beta^* = 1.05$.

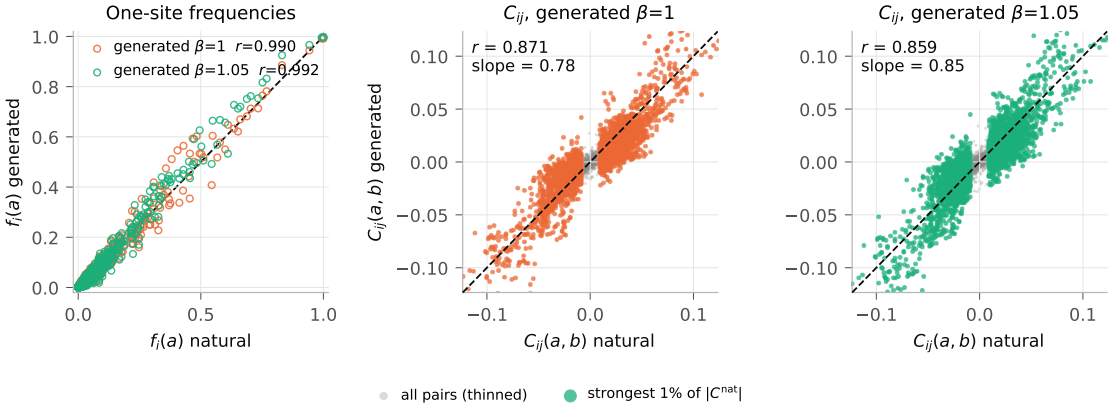


Figure 4: Generated vs. natural (reweighted test) one-site frequencies and connected correlations. The C_{ij} panels show the strongest 1% of $|C^{\mathrm{nat}}|$ in colour over a thinned grey sample of all 861 273 entries. Dashed: identity.

statistic	set vs. natural test	Pearson r	Spearman ρ	slope	RMSE
$f_i(a)$	generated, $\beta = 1$	0.990	0.972	1.01	0.0160
	generated, $\beta^* = 1.05$	0.992	0.970	1.06	0.0165
	natural train (ceiling)	0.998	0.983	0.99	0.0070
	independent-site model	0.998	0.982	0.99	0.0072
$C_{ij}(a,b)$	generated, $\beta = 1$	0.871	0.193	0.78	0.0015
	generated, $\beta^* = 1.05$	0.859	0.209	0.85	0.0016
	natural train (ceiling)	0.944	0.377	0.88	0.0010
	independent-site model	-0.005	0.058	-0.00	0.0030
C_{ij}, top 1%	generated, $\beta = 1$	0.918	0.880	0.84	0.0113
	generated, $\beta^* = 1.05$	0.908	0.865	0.93	0.0123
	natural train (ceiling)	0.981	0.949	0.91	0.0057
	independent-site model	-0.009	-0.022	-0.00	0.0283

Table 2: Agreement with the natural test set (natural sets reweighted, generated uniform). "Top 1%": the 1% largest $|C^{\mathrm{test}}_{ij}(a,b)|$.

4 Discussion

What the attention parametrisation buys. With 217 728 parameters — a quarter of the 861 273 entries of a free coupling tensor — the model reaches near-perfect contact prediction and a correlation agreement within a few percent of the finite-sample ceiling over 85.9% of the family. The factorisation is not merely a compression: sharing V^h across all position pairs is what makes the effective number of parameters small enough for $M_{\mathrm{eff}} \approx 3590$ sequences to constrain.

The β offset is a property of pseudo-likelihood, not a bug. Maximising Eq. (4) fixes each conditional but leaves the overall scale of E unconstrained, so there is no reason for the $\beta = 1$ chain to sit at the natural energy. The scan of Fig. 3 shows the two failure modes on either side: too low a β gives over-dispersed, weakly correlated samples, too high a β collapses diversity. Matching the mean energy of the training set picks $\beta^* = 1.05$ without reference to held-out data and recovers the correlation amplitude almost exactly.

The missing mode is the substantive limitation. The under-sampled HscB-like group is not a sampling artefact — chains started inside it leave — but a statement about the learned landscape: that basin is 14 energy units too shallow, worth +1.01 nats of free energy. The diagnosis in Table 3 points at the fix, because the failure is in the prior rather than in the condition-

cluster	weight share		$\langle E \rangle$ natural		$\langle \log \mathrm{PL} \rangle$ natural		ΔF
	natural	generated	test	train	test	train	(nats)
C0†	0.141	0.051	-119.5	-122.6	-80.5	-71.8	+1.01
C1	0.387	0.352	-134.0	-134.5	-76.8	-71.1	+0.09
C2	0.473	0.597	-133.5	-137.2	-79.7	-71.0	-0.23

Table 3: Per-cluster diagnosis ($\beta = 1$); natural averages use the reweighting weights. $\Delta F = -\ln(s^{\mathrm{gen}}/s^{\mathrm{nat}})$ is the free-energy error in nats. † marks the missed cluster: its conditionals are fitted as well as anywhere, but its energy is far higher.

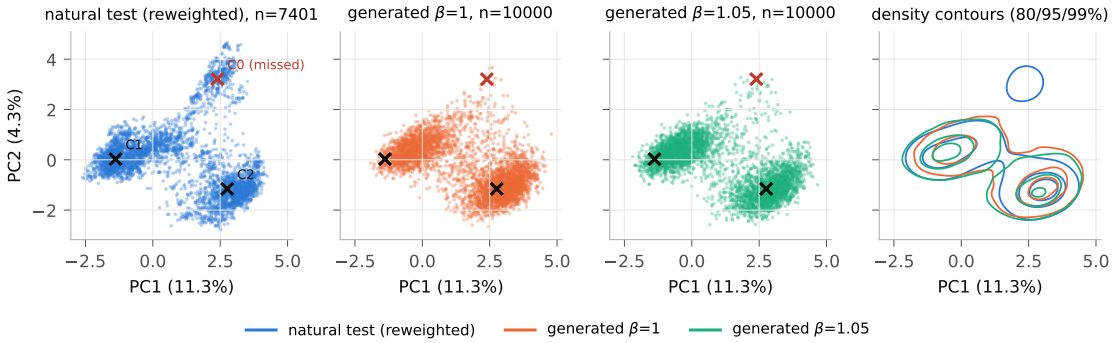


Figure 5: Test and generated sequences on the first two principal components of the one-hot test alignment (PCA fitted on test only); each panel draws a random sample, the natural panel drawn with its reweighting weights. Crosses: k-means centres, red for the missed cluster.

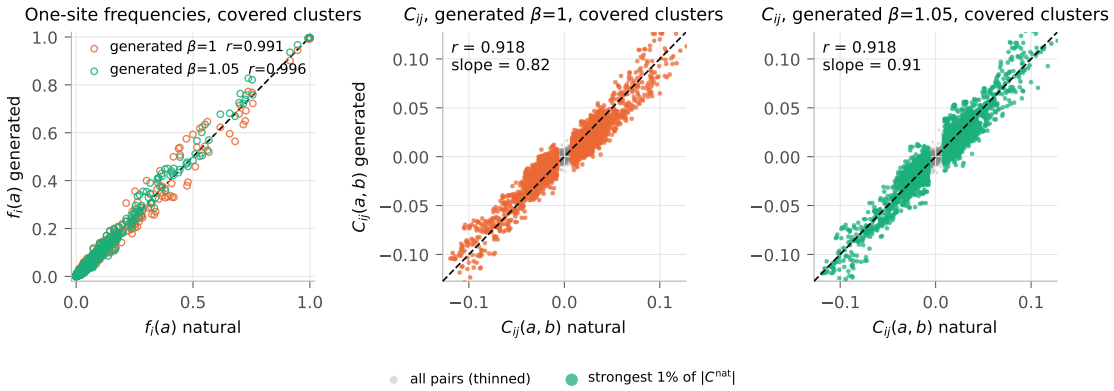


Figure 6: As Fig. 4, with natural and generated sets both restricted to the covered PCA clusters.

als. Adding explicit fields h_i, fitting a mixture over sub-families, or reweighting the training set towards the minority group would each give the model somewhere to put that mass; sampling from a mixture of temperatures would paper over it, since Fig. 3 shows the minor cluster is relatively better populated at lower β. All are single-parameter experiments on top of the existing pipeline.

On evaluating generative sequence models. Two methodological points generalise beyond this family. First, a species-disjoint split is not a sequence-disjoint split; had we not audited the files, the held-out likelihood would have looked 11 percentage points better than it is. Second, aggregate correlation scores conflate "distorts the covariation" with "misses a sub-family". Partitioning the family and re-scoring inside the covered modes separates the two, and here it moves the interpretation from "the model captures 87% of the correlation structure" to "the model is near the sampling limit on 85.9% of the family and misses the rest". Those are different problems with different fixes.

5 Conclusion

An attention-parametrised Potts model trained by pseudo-likelihood on 29 312 bacterial J-domains generates sequences that are new, as diverse as natural ones, and statistically faithful over the bulk of the family: $r = 0.990$ on site frequencies, $r = 0.871$ on connected correlations, rising to 0.918 against a 0.941 ceiling on the modes the sampler populates. The residual error is concentrated in one under-weighted sub-family rather than spread over the covariation structure, and the sampler itself is fast ($\sim$0.3 μs per chain-update) and well-mixed ($\tau_{\mathrm{int}} = 6.5$ sweeps).

Code: `attdca/` (package), `scripts/` (pipeline), `tests/` (vectorised likelihood, energy, sampler and split checked against naive loops and an exact two-site Gibbs target). A longer companion report documents the derivations and the full metric definitions.

statistic	set vs. natural (covered clusters)	Pearson r	Spearman ρ	slope	RMSE
$f_i(a)$	generated, $\beta = 1$	0.991	0.970	0.98	0.0153
	generated, $\beta^* = 1.05$	0.996	0.969	1.03	0.0118
	natural train (ceiling)	0.998	0.980	0.98	0.0079
	independent-site model	0.993	0.973	0.95	0.0144
$C_{ij}(a, b)$	generated, $\beta = 1$	0.918	0.179	0.82	0.0012
	generated, $\beta^* = 1.05$	0.918	0.199	0.91	0.0013
	natural train (ceiling)	0.941	0.345	0.87	0.0010
	independent-site model	-0.013	0.060	-0.00	0.0031
C_{ij}, top 1%	generated, $\beta = 1$	0.961	0.919	0.87	0.0084
	generated, $\beta^* = 1.05$	0.961	0.919	0.97	0.0083
	natural train (ceiling)	0.979	0.938	0.91	0.0061
	independent-site model	-0.031	-0.035	-0.00	0.0296

Table 4: Statistics restricted to the covered PCA clusters (85.9% of the natural test weight). Compare Table 2.

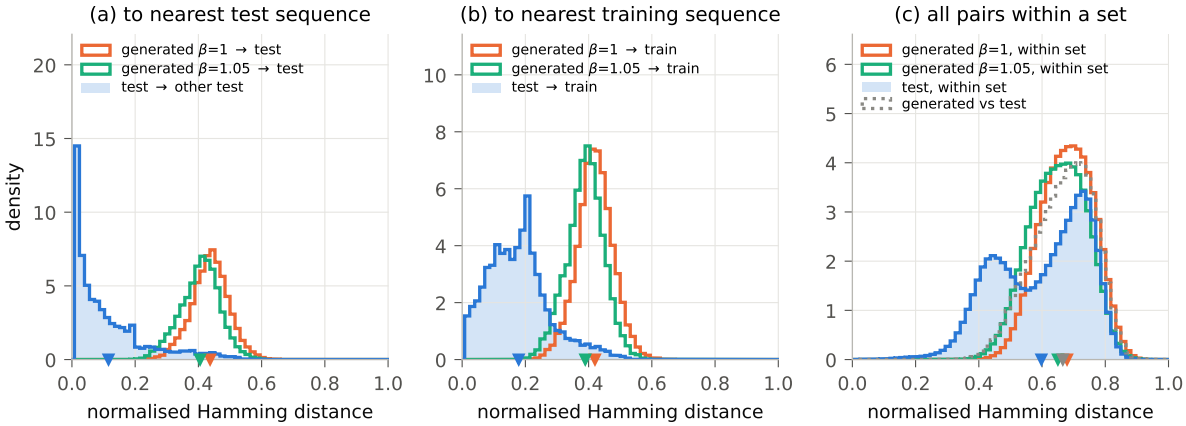


Figure 7: Normalised Hamming distances; triangles mark means. The blue reference curve is different in each panel. **(a)** distance from each sequence to the nearest sequence *of the natural test set*; blue is a test sequence to its nearest *other* test sequence (leave-one-out), i.e. how close natural relatives normally lie — its spike near zero is the family's phylogenetic redundancy. **(b)** distance to the nearest *training* sequence, the memorisation check; blue is the held-out test set measured identically, the closest a genuine unseen J-domain comes to the training data. **(c)** all pairwise distances *within* one set (3000 sequences each), i.e. the spread of the set itself; blue is the natural test set and the dotted grey curve is generated-vs-test, which coincides with the generated curve when both sets occupy the same region of sequence space.

distance (normalised by L)	mean	median	5% quantile	$= 0$
gen ($\beta = 1$) → nearest test	0.437	0.444	0.333	0.0%
gen ($\beta = 1$) → nearest train	0.420	0.413	0.317	0.0%
gen ($\beta = 1$) → nearest other gen	0.430	0.429	0.349	0.0%
gen–gen, all pairs ($\beta = 1$)	0.679	0.683	0.540	0.0%
gen–test, all pairs ($\beta = 1$)	0.666	0.683	0.492	0.0%
gen ($\beta^* = 1.05$) → nearest test	0.405	0.413	0.302	0.0%
gen ($\beta^* = 1.05$) → nearest train	0.389	0.397	0.286	0.0%
gen ($\beta^* = 1.05$) → nearest other gen	0.394	0.397	0.317	0.0%
gen–gen, all pairs ($\beta^* = 1.05$)	0.650	0.651	0.508	0.0%
gen–test, all pairs ($\beta^* = 1.05$)	0.639	0.651	0.460	0.0%
test → nearest other test	0.116	0.079	0.016	0.0%
test → nearest train	0.179	0.175	0.032	0.0%
novel test → nearest train	0.179	0.175	0.032	0.0%
test–test, all pairs	0.598	0.635	0.365	0.0%

Table 5: Hamming-distance summary; "= 0" is the fraction of exact copies.